Currarino Triad (Currarino Syndrome) — Comprehensive Disease Characteristics Report

Disease: Currarino Triad / Currarino syndrome (ASP triad) **MONDO:** MONDO:0008305 · **OMIM:** #176450 · **Orphanet:** ORPHA:1552 **ICD-10:** Q87.8 (approximate; often coded via component anomalies Q06.8/Q76.49) · **ICD-11:** component-based (LD2F.1Y / LB70.Y) **MeSH:** "Currarino syndrome" (via Digestive System Abnormalities; Sacrococcygeal Region) **Causal gene: MNX1** (motor neuron and pancreas homeobox 1; formerly **HLXB9**), HGNC:4628, located at **chromosome 7q36.3 Category:** Mendelian, autosomal dominant

Summary

Currarino syndrome (CS) is a rare, congenital, autosomal-dominant caudal malformation syndrome caused by heterozygous loss-of-function variants in the homeobox transcription-factor gene MNX1 (formerly HLXB9) at chromosome 7q36.3. Through haploinsufficiency, loss of one functional MNX1 allele disrupts dorsal–ventral patterning of the developing caudal (hindgut/notochord/neural) region, producing the defining clinical **triad**: (1) partial sacral agenesis with a characteristic sickle-shaped or "scimitar" hemisacrum, (2) an anorectal malformation — most commonly anal stenosis — and (3) a presacral mass (usually an anterior meningocele, a mature teratoma, or both). The triad is frequently incomplete, and associated anomalies (tethered cord, urogenital/gynecologic malformations, renal anomalies, recurrent meningitis) are present in roughly 85% of patients.

The disorder is defined by **incomplete, age-independent penetrance** (~50% of MNX1 mutation carriers are clinically asymptomatic) and **highly variable expressivity**, with no reproducible genotype–phenotype correlation even within families sharing the same variant. MNX1 pathogenic variants are detected in nearly all **familial** cases but only in approximately **25–30% of sporadic** cases, implying additional genetic/modifier contributions in mutation-negative patients. The variant spectrum is dominated by null alleles (frameshift, nonsense,

splice-site, whole-gene/7q36 deletions) with mutational hotspots in exon 1 (the N-terminal polyalanine tract) and the homeobox DNA-binding domain; mosaic mutations and translocation-mediated 7q36 deletions are also reported.

Diagnosis rests on clinical suspicion (persistent childhood constipation + sacral anomaly), **MRI** (the most sensitive single modality for presacral mass, spinal dysraphism, and anorectal malformation), and **MNX1 sequencing plus deletion/MLPA/chromosomal-microarray testing**. Management is **multidisciplinary surgery** — presacral mass resection, spinal cord untethering, and anorectoplasty — most often via posterior/transsacral or combined anterior–posterior approaches. Prognosis is generally good because CS-associated presacral masses are predominantly **benign mature teratomas**, but chronic constipation (~76%) and urinary incontinence (~44%) are common long-term morbidities, and rare malignant (neuroendocrine) transformation warrants lifelong surveillance. There is no pharmacologic or gene therapy; prevention centers on cascade family screening and genetic counseling. The principal experimental model is the **Hlxb9/Mnx1-null mouse**, and MNX1 forms an **allelic series** in which biallelic missense variants instead cause neonatal diabetes and overexpression drives certain cancers.

Key Findings

Finding 1 — Causal gene and variant spectrum: heterozygous loss-of-function MNX1 at 7q36

Currarino syndrome is caused by **heterozygous loss-of-function mutations in MNX1 (HLXB9)** on chromosome 7q36.3. MNX1 encodes HB9, a homeodomain transcription factor. The detection rate differs sharply by inheritance context: pathogenic variants are found in **nearly all familial CS cases** but only **~28–30% of sporadic cases**. As stated by PMID: 34156759: *"CS is associated with mutations in the MNX1 gene (motor neuron and pancreas homeobox-1, previously known as HLXB9) mapped to chromosome 7q36. Heterozygous loss-of-function mutations in the coding sequence of MNX1 gene have been reported in nearly all familial CS cases and in approximately 30% of CS sporadic patients."* The Korean cohort of PMID: 29401559 quantified this precisely: *"Pathogenic variants in MNX1 were detected in 28% of all CS cases and 25% of sporadic cases."*

The variant spectrum is dominated by predicted null alleles consistent with haploinsufficiency:

Variant type	Representative examples
Frameshift	p.Pro18ProfsX38, p.Gly98Alafs*124*, *p.Gly145Alafs*77, p.Gly151Leufs*67*, *p.Ala216Profs*5
Nonsense	p.Gln212, *p.Tyr186*, p.Gln288X
Missense (homeobox)	p.Arg243Trp, p.Lys295Gln, p.Trp288Gly
Splice-site	IVS2+1G>A
Structural	Large 7q36 deletions; whole-gene deletions; der(7)t(7;21)(q36.2;p11.3)
Mosaic	Reported in an adult female with CS features

The Swedish series (PMID: 21763840) demonstrated both de novo (p.Gln212X) and maternally inherited (p.Pro18ProfsX38) variants, and showed that MLPA and array-CGH are required to capture deletions missed by sequencing. **Suggested ontology terms:** gene HGNC:4628 (MNX1); disease MONDO:0008305.

Finding 2 — MNX1/HB9 developmental role (mouse and in vitro)

MNX1/Hlxb9 is a developmental master regulator required for caudal endodermal and neural patterning. In the mouse knockout (PMID: 10471502): *"In mice homologous for a null mutation of Hlxb9, the dorsal lobe of the pancreas fails to develop. The remnant Hlxb9-/- pancreas has small islets of Langerhans with reduced numbers of insulin-producing beta-cells."* The gene is also essential in the nervous system: PMID: 25136833 notes HB9 *"is relevant for both motor neuronal and pancreatic beta cells development"* and that *"Experimental work on animal models has shown that HLXB9 has an essential role in motor neuronal differentiation."* This dual (endoderm + neuroectoderm) requirement mechanistically links MNX1 loss to the combined hindgut/sacral/neural tube phenotypes seen in CS. **Suggested GO terms:** GO:0030154 (cell differentiation), GO:0021520 (spinal cord motor neuron cell fate specification), GO:0031016 (pancreas development).

Finding 3 — Presacral mass: predominantly benign teratoma/meningocele, rare malignancy

CS presacral masses are biologically distinct from sporadic sacrococcygeal teratomas (SCTs). In a multicenter review of 203 pediatric SCT resections (2010–2020), 25 (12.3%) had CS; **all CS tumors were mature teratomas** (PMID: 40541662): *"All tumors in the Currarino cohort were*

mature teratomas, while 27 % of non-Currarino tumors were immature and 12 % were malignant (p < 0.001). Tumors in Currarino patients were significantly smaller in size (median 3.3 cm vs. 8.0 cm, p < 0.001)." CS tumors were predominantly Altman type IV (87% vs 17%), more often diagnosed postnatally (80% vs 25%), and associated with higher urinary incontinence (44% vs 28%, p=0.048) and constipation (76% vs 32%, p<0.001).

Despite the benign norm, malignant transformation occurs rarely. PMID: 21915987: *"Since malignant neuroendocrine transformation of presacral mass in CS is a possible complication, even thought rare, close follow up in these patients is advisable."* The mass may also be an anterior meningocele, a dermoid/neurenteric/enterogenous cyst, a hamartoma, or **dual pathology** — in a review of ~200 complete-triad cases, 22 had both a meningocele and a teratoma (PMID: 12847376).

Finding 4 — Incomplete penetrance and absence of genotype–phenotype correlation

CS shows strikingly variable expressivity and reduced penetrance. In a study of 72 individuals from 9 families (PMID: 11528505): *"Highly variable phenotypes and a low penetrance with half of all carriers being clinically asymptomatic were found in three families, whereas affected members of one family showed almost identical phenotypes. However, an obvious genotype-phenotype correlation was not found."* The same work localized mutational hotspots: *"The distribution pattern of here and previously reported HLXB9 mutations indicates mutational predilection sites within exon 1 and the homeobox."* The ~50% asymptomatic-carrier figure is the single most important fact for genetic counseling and cascade screening — an apparently "sporadic" child may have a carrier parent harboring a presacral mass. The siblings in PMID: 23466002 with an identical HLXB9 mutation showed phenotypic expression ranging from incomplete to complete triad, reinforcing variable expressivity.

Finding 5 — MRI is the preferred modality; frequent coexisting spinal dysraphism

Among 32 pediatric complete-CS patients, MRI outperformed CT for most features (PMID: 35282111): *"While CT was better than MRI in displaying sacral anomaly (P<0.05), MRI was more sensitive than CT at detecting presacral mass, spinal dysraphism, and congenital anal atresia (P<0.05)."* In that cohort, anal atresia was the most common ARM (59.4%), sacral agenesis was mainly type IV (75%), and 25/32 had associated anomalies (tethered cord, filum lipoma, hydronephrosis). A surgical series of 10 CS patients (PMID: 30095346) documented the dysraphism spectrum: *"Spinal anomalies included anterior meningocele (5 cases), low-lying*

conus (8), terminal syrinx (4), fatty filum (5), caudal lipoma (3), and intraspinal cyst (1)." The pathognomonic bony lesion is the **sickle-shaped / "scimitar" hemisacrum** (preserved S1 with asymmetric defect of lower sacrum).

Finding 6 — Anterior sacral meningocele: female predominance, surgical approaches

Anterior sacral meningocele (ASM) is the most common presacral-mass type. A systematic review of 208 ASM patients across 148 studies (PMID: 42508731) found: *"Associated malformative syndromes were identified in 66% of cases, most commonly Currarino syndrome (33%). The dorsal transsacral approach was the most frequently performed procedure (58%) and showed favorable outcomes."* Female predominance was 73% (mean age 24.6 years); constipation was the most common presenting symptom (50%). An independent case series/review (PMID: 41989599) confirmed 81% female predominance and a 70% preference for posterior approaches with lower morbidity/recurrence. Untreated ASM can present acutely as bowel obstruction/ileus or recurrent meningitis; CS can present in the neonatal period with obstruction (PMID: 22901924): *"We describe a male infant presenting in the neonatal period with bowel obstruction who had features of anal stenosis, a presacral teratoma, and a sacral anomaly consistent with Currarino's syndrome."*

Finding 7 — Phenotypic stratification (Complete / Mild / Minimal) and high associated-anomaly rate

A multicentre retrospective study of 45 CS patients (PMID: 28689883) defined three phenotypic subtypes: *"Twenty patients (44.5%) presented a Complete CS type, 19 (42.2%) a Mild CS and 6 (13.3%) a Minimal CS. In addition to the classical triad elements, 38 (84.5%) patients showed associated anomalies."* Critically, this study also addressed the mutation-negative problem: *"Mutations in the MNX1 gene are the main genetic background of CS, although they are not present in almost half of the cases."* Associated anomalies include tethered cord, urogenital/gynecologic anomalies (bicornuate uterus, vaginal/uterine duplication), renal anomalies (hydronephrosis), and recurrent meningitis.

Finding 8 — Principal model organism: the Hlxb9/Mnx1-null mouse

The Hlxb9-null mouse (PMID: 10471502) is the principal experimental model, recapitulating MNX1 developmental dependence through dorsal pancreatic lobe agenesis and reduced beta-cell numbers. HB9 is also essential for spinal motor neuron differentiation (PMID: 25136833), and the human SK-N-BE neuroblastoma line models HLXB9/HB9 transcriptional dynamics

during retinoic-acid-induced neuronal differentiation. **Key limitation:** the mouse null phenotype centers on the pancreas/beta-cells, whereas human CS centers on caudal sacral/anorectal/presacral features; moreover, human CS is a heterozygous haploinsufficiency disorder, while the informative mouse phenotype is homozygous-null. The mouse ortholog is *Mnx1* (MGI). No naturally occurring animal equivalent is recorded in OMIA.

Finding 9 — HB9 is a nuclear transcription factor interacting with NONO; biallelic MNX1 is a distinct allelic disorder

HB9 is a nuclear homeodomain protein whose phosphorylated isoform interacts with NONO/p54NRB (PMID: 29309627): *"the phosphorylated isoform of the β-cell differentiation factor HLXB9 that interacts with NONO/p54NRB, a survival factor."* Importantly, this establishes an **allelic series**: biallelic/homozygous missense mutations cause a recessive disorder distinct from CS — *"two different homozygous germline mutations in HLXB9, p.F248L and p.F272L, were reported in neonatal diabetes, a condition with functional β-cell deficiency."* Separately, MNX1 **overexpression** is oncogenic (e.g., bladder cancer via CCNE1/2 upregulation PMID: 30012177; infant leukemia via t(7;12)). Thus MNX1 dosage maps to three distinct disease states.

Finding 10 — Treatment-resistant anal stenosis; broad presacral-mass differential

CS-associated anal stenosis behaves differently from isolated congenital anal stenosis. In a systematic review of 70 congenital anal-stenosis cases (PMID: 42489929): *"20 cases (28.6%) associated with Currarino syndrome showed poor response to conservative treatment and frequently required treatment of associated presacral lesions or more invasive procedures."* The presacral-mass differential extends beyond meningocele/teratoma (PMID: 42864507): *"Final pathologic diagnoses included myxopapillary ependymoma, mature cystic teratoma, and neuroendocrine tumor arising within a teratoma. These cases highlight the considerable heterogeneity of presacral lesions in Currarino syndrome."*

Finding 11 — Overall disease synthesis

CS is a congenital autosomal-dominant caudal malformation triad: partial sacral agenesis + anorectal malformation + presacral mass, caused by heterozygous MNX1/HLXB9 LoF at 7q36 (AD, ~50% asymptomatic carriers, variable expressivity). Diagnosis combines clinical findings, MRI, and MNX1 sequencing/MLPA/CMA. Treatment is surgical and multidisciplinary; prognosis is generally good (predominantly benign mature teratomas, 0% malignant in one 25-patient CS

cohort vs 12% in non-CS SCT), though chronic constipation (~76%) and urinary incontinence (~44%) are common long-term morbidities (PMID: 40541662; PMID: 34156759). Orphanet prevalence <1/100,000.

Detailed Section Report

1. Disease Information

CS is a rare congenital malformation syndrome of the caudal body region, first delineated as a triad by Currarino et al. (1981). **Identifiers:** MONDO:0008305; OMIM #176450; Orphanet ORPHA:1552; MeSH "Currarino syndrome." **Synonyms:** Currarino syndrome, Currarino triad, ASP triad (Anorectal malformation–Sacral defect–Presacral mass), hereditary sacral agenesis with presacral mass. Information is primarily **disease-level aggregated** from case reports, case series, and multicenter cohorts rather than large EHR datasets, reflecting the disorder's rarity.

2. Etiology

Primary cause: heterozygous loss-of-function MNX1 variants (haploinsufficiency). **Genetic risk factors:** the MNX1 variant itself; a positive family history is the strongest risk factor (familial transmission with ~50% penetrance). Modifier genes are hypothesized because mutation-negative cases (~50% of sporadic patients) and intrafamilial variability imply additional genetic background (PMID: 19853743: *"The variability of the CS phenotype among related or unrelated patients bearing the same mutation advocates for differences in the genetic background of each individual and invokes the implication of additional CS susceptibility genes."*). **Environmental risk/protective factors:** none established — this is a Mendelian developmental disorder. **Gene–environment interactions:** not documented. **Protective factors:** none known.

3. Phenotypes

Phenotype	Type	HPO term (suggested)	Onset	Frequency
Partial sacral agenesis / scimitar hemisacrum	Skeletal sign (imaging)	HP:0008436 (Abnormal sacrum morphology)	Congenital	Triad-defining (~all complete cases)
Anorectal malformation / anal stenosis	Structural sign	HP:0002025 (Anal stenosis) / HP:0002023 (Anal atresia)	Congenital/ neonatal	Most common ARM; anal atresia 59.4% in one cohort
Presacral mass (meningocele/ teratoma)	Structural sign	HP:0032417 (Presacral mass / teratoma terms)	Congenital	Triad-defining
Chronic constipation	Symptom	HP:0002019	Infancy–childhood	~76%
Urinary incontinence	Symptom	HP:0000020	Variable	~44%
Tethered cord	Neurological sign	HP:0002144	Congenital	Frequent (part of ~85% associated anomalies)
Recurrent meningitis	Clinical sign	HP:0002383	Variable	Associated (via meningocele)
Urogenital/ gynecologic anomalies	Structural sign	HP:0000130 / HP:0000078	Congenital	Associated

Severity/progression: variable; many carriers asymptomatic (mild/minimal subtypes). Symptoms are generally stable/structural but complications (obstruction, infection) may be acute. **Quality-of-life impact:** chronic constipation and incontinence are the dominant long-term burdens.

4. Genetic/Molecular Information

Causal gene: MNX1/HLXB9 (HGNC:4628; OMIM *142994), 7q36.3. Variant classes: frameshift, nonsense, splice-site, missense (homeobox), whole-gene/7q36 deletions, mosaicism (see Finding 1 table). Classification: most null variants are Pathogenic/Likely Pathogenic per ACMG (PVS1 loss-of-function mechanism). Allele frequency: pathogenic CS variants are essentially absent from gnomAD (private/family-specific). Origin: germline (de novo or inherited); one mosaic case reported. Functional consequence: loss of function / haploinsufficiency; homeobox missense variants may alter DNA-binding specificity. Modifier genes: hypothesized but unidentified. Epigenetics: no disease-specific methylation signature established. Chromosomal abnormalities:* 7q36 deletions, including translocation-mediated der(7)t(7;21)(q36.2;p11.3) (PMID: 38585544).

5. Environmental Information

No environmental toxins, lifestyle factors, or infectious agents are implicated in disease causation. Recurrent meningitis is a *complication* (via CSF-communicating meningocele), not an etiologic agent.

6. Mechanism / Pathophysiology

Ordered causal chain:

1. A **heterozygous loss-of-function MNX1 variant** (frameshift/nonsense/splice/deletion, or rarely a DNA-binding missense) → reduces functional HB9 transcription factor to ~50% (**haploinsufficiency**).
2. HB9 haploinsufficiency → **impaired transcriptional regulation of caudal developmental patterning** during early embryogenesis (inferred from the gene's known role in motor neuron and endodermal/beta-cell development; a direct caudal-patterning assay in human CS has not been demonstrated).
3. Defective patterning → **disrupted dorsal–ventral/notochord–hindgut–neural tube coordination** in the caudal eminence (inferred — CS is described as caused by dorsal-ventral patterning defects during embryonic development).
4. This **branches** into the triad:
5. 4a → **partial sacral agenesis** (scimitar hemisacrum) via failed caudal vertebral/sacral formation;
6. 4b → **anorectal malformation** (anal stenosis/atresia) via abnormal hindgut/cloacal development;

7. 4c → **presacral mass** (anterior meningocele from dural herniation through the sacral defect, and/or teratoma from persistent caudal pluripotent cells).
8. Structural lesions → **clinical manifestations**: constipation/obstruction (ARM + mass effect), tethered cord/neurological deficit, recurrent meningitis (CSF-communicating meningocele), urinary/fecal incontinence.
9. Rare downstream branch: persistent germ-cell/neuroendocrine elements in the teratoma → **malignant transformation** (neuroendocrine tumor) — rare.

Molecular pathways/processes: MNX1/HB9 is a homeodomain transcription factor acting in embryonic developmental gene-regulatory networks; its phosphorylated isoform interacts with **NONO/p54NRB** (PMID: 29309627). **Protein dysfunction:** loss of function (truncation) or altered DNA binding (homeobox missense). **Cell types involved (suggested CL):** motor neurons (CL:0000100), pancreatic beta cells (CL:0000169) in the broader MNX1 biology; caudal mesenchyme/notochord and hindgut endoderm in CS proper. **GO terms:** GO:0045944 (positive regulation of transcription by RNA Pol II), GO:0009952 (anterior/posterior pattern specification), GO:0048646 (anatomical structure formation in morphogenesis). The upstream lesion is the mutation; all structural anomalies are downstream and parallel (the triad branches share a common upstream patterning defect rather than forming a linear cascade among themselves).

7. Anatomical Structures Affected

Primary organs/structures (UBERON): sacrum (UBERON:0001684), anus/rectum/anal canal (UBERON:0001052/UBERON:0001353), presacral/retrorectal space. **Secondary involvement:** spinal cord and meninges (tethered cord, meningocele; UBERON:0002240/UBERON:0002360), urinary tract (hydronephrosis; UBERON:0000056), female genital tract (uterus/vagina duplications; UBERON:0000995/UBERON:0000996). **Body systems:** digestive, nervous (including caudal spinal cord), skeletal, genitourinary. **Tissue level:** neuroectoderm (meninges/cord), endoderm (hindgut), mesoderm (sacrum), plus germ-cell-derived teratoma tissue. **Subcellular (GO CC):** nucleus (GO:0005634) — HB9 is a nuclear transcription factor. **Lateralization:** the hemisacral defect is characteristically **asymmetric/unilateral** (scimitar).

8. Temporal Development

Onset: congenital; anomalies are present at birth. **Presentation:** often in infancy/childhood (constipation, sacral dimple, obstruction), but frequently diagnosed in **adulthood** or incidentally (e.g., ASM found during pregnancy/delivery evaluation) because of mild/minimal phenotypes. **Course:** structural and stable rather than progressive, but complications

(obstruction, meningitis, rare malignant transformation) can be acute/episodic. **Duration:** lifelong. **Critical period:** early embryogenesis (caudal development) is the window of vulnerability; postnatally, early surgical diagnosis prevents morbidity.

9. Inheritance and Population

Inheritance: autosomal dominant. **Penetrance:** incomplete — ~50% of carriers asymptomatic (PMID: 11528505). **Expressivity:** highly variable; no genotype–phenotype correlation. **Mosaicism:** reported (somatic/germline mosaic MNX1). **Founder effects:** none established. **Consanguinity:** not a major factor for dominant CS (relevant instead to the recessive allelic neonatal diabetes). **Epidemiology:** rare; Orphanet prevalence <1/100,000. **Sex ratio:** overall ~equal for CS, but the ASM subset shows strong **female predominance (73–81%)**. **Geographic distribution:** worldwide; cases reported across European, East Asian, and North African populations.

10. Diagnostics

Imaging: MRI is the most sensitive single modality for presacral mass, spinal dysraphism, and ARM; CT better depicts the bony sacral defect (PMID: 35282111). Plain radiograph may show the scimitar hemisacrum. **Genetic testing (recommended approach):** MNX1 single-gene sequencing first; if negative, **MLPA/deletion analysis and chromosomal microarray** to detect 7q36 deletions (PMID: 21763840); karyotype/FISH where translocation suspected. WES/WGS useful in syndromic/mutation-negative cases. **Clinical criteria:** the triad (sacral defect + ARM + presacral mass), with "incomplete triad" recognized. **Differential diagnosis:** caudal regression syndrome (broader caudal agenesis; HLXB9 **not** involved — PMID: 16498628), isolated sacrococcygeal teratoma, VACTERL association, isolated anorectal malformation, Hirschsprung disease (which can co-occur — PMID: 23948812). **Screening:** MRI screening of at-risk family members of a known mutation carrier is recommended to detect occult presacral masses (PMID: 23466002).

11. Outcome / Prognosis

Prognosis is **generally good**: CS presacral masses are predominantly benign mature teratomas (0% malignant in one 25-patient CS cohort vs 12% malignant in non-CS SCT; PMID: 40541662). **Morbidity** is dominated by chronic constipation (~76%) and urinary incontinence (~44%), plus risk of recurrent meningitis and, rarely, neuroendocrine malignant transformation requiring

lifelong surveillance (PMID: 21915987). Life expectancy is near-normal with appropriate management. **Prognostic factors:** completeness of triad, extent of spinal/neurological involvement, timeliness of surgery, and mass histology.

12. Treatment

No pharmacologic or gene therapy exists; management is **surgical and supportive/multidisciplinary** (pediatric surgery + neurosurgery). **Surgical interventions (suggested NCIT):** resection of the presacral mass (posterior sagittal / transsacral / anterior transabdominal / combined approaches), spinal cord **untethering**, **anorectoplasty** (e.g., PSARP) for the ARM, and **colostomy** for acute obstruction/neonatal management (PMID: 30095346; PMID: 23590527). For ASM, posterior/dorsal transsacral approaches are most frequent and lowest-morbidity; combined anterior–posterior approaches have highest success for large/high-lying lesions (PMID: 42508731; PMID: 41989599). CS-associated anal stenosis often resists conservative dilatation and requires definitive surgery (PMID: 42489929). **Supportive care:** bowel management for constipation, management of urinary dysfunction, antibiotics for meningitis. Anesthetic planning is important given sacral/dural anomalies (neuraxial anesthesia challenges — PMID: 41916044).

13. Prevention

Primary prevention is not possible (congenital Mendelian disorder). **Secondary prevention:** cascade **family/MRI screening** of first-degree relatives of mutation carriers to detect occult masses early; prenatal/postnatal imaging. **Genetic counseling** (AD inheritance, 50% transmission risk but reduced penetrance) and options for **prenatal testing / preimplantation genetic diagnosis** where a familial variant is known. **Tertiary prevention:** surgical correction and surveillance to prevent complications (obstruction, meningitis, malignant transformation).

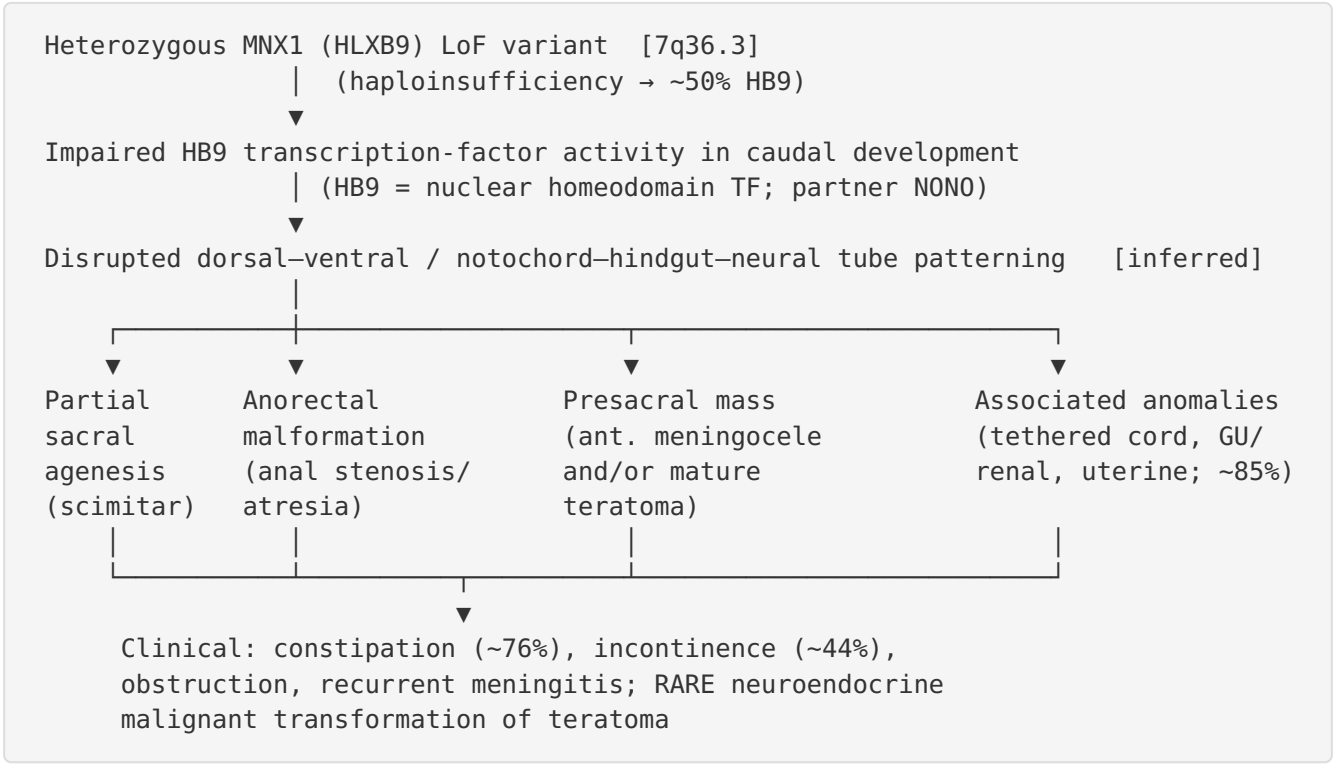
14. Other Species / Natural Disease

Taxonomy: human (NCBI Taxon 9606) disease; experimental mouse (*Mus musculus*, 10090) model. **Orthologous gene:** mouse *Mnx1* (MGI). **Natural disease:** no naturally occurring CS equivalent is recorded in OMIA; no zoonotic or cross-species transmission (non-infectious genetic disorder). Evolutionary conservation of MNX1/HB9's developmental role (motor neurons, pancreas) across vertebrates underpins model-organism relevance.

15. Model Organisms

Principal model: the **Hlxb9/Mnx1-null mouse** (knockout), which recapitulates MNX1 developmental dependence via dorsal pancreatic lobe agenesis and reduced beta-cell numbers (PMID: 10471502), and informs HB9's role in motor-neuron differentiation (PMID: 25136833). **In vitro:** human SK-N-BE neuroblastoma cells model HLXB9/HB9 transcriptional dynamics during RA-induced differentiation; MIN6 beta-cell line used for NONO-interaction and proliferation studies. **Limitations:** the mouse null phenotype is pancreas/beta-cell–centric and homozygous, whereas human CS is a heterozygous, caudal-skeletal/anorectal/presacral disorder — the model therefore captures MNX1 molecular biology better than the human malformation phenotype. **Resources:** MGI (mouse), IMPC/IMSR for alleles.

Mechanistic Model (Schematic)



MNX1 dosage → three distinct diseases (allelic series):

MNX1 state	Mechanism	Disease
Heterozygous LoF	Haploinsufficiency	**Currarino syndrome** (caudal malformation)
Biallelic/homozygous missense (p.F248L, p.F272L)	β-cell dysfunction	Neonatal diabetes
Overexpression	Oncogenic (CCNE1/2↑; t(7;12))	Bladder cancer; infant leukemia

Evidence Base

PMID	Title (abbrev.)	Supports
34156759	Mosaic MNX1 in adult female CS	Causal gene, locus, detection rates
29401559	MNX1 variant spectrum, Korean CS	28%/25% detection; variant classes
10471502	Hlxb9-null mice	Developmental role; mouse model
25136833	HLXB9 in neuronal differentiation	Motor-neuron + beta-cell role
40541662	SCT in CS, multicenter	Benign teratoma predominance; outcomes
21915987	CS with neuroendocrine tumor	Rare malignant transformation
11528505	Genotype–phenotype analysis	~50% penetrance; mutational hotspots
35282111	CT vs MRI in CS	MRI superiority; phenotype frequencies
30095346	Combined surgical repair	Spinal dysraphism spectrum; surgery
28689883	Multicentre CS subtypes	Complete/Mild/Minimal; 84.5% associated
42508731	ASM surgical review	CS 33% of ASM; approaches
41989599	ASM case series/review	Female predominance; posterior approach
29309627	HLXB9/NONO functional defects	NONO interaction; neonatal-diabetes allele
42489929	Congenital anal stenosis review	Treatment-resistant CS stenosis
42864507	Atypical presacral masses	Broader mass differential
19853743	MNX1 mutations in CS patients	Modifier/susceptibility-gene hypothesis
21763840	Swedish MNX1 analysis	MLPA/CGH utility; asymptomatic carriers
16498628	HLXB9 & caudal regression	DDx: HLXB9 not causal in CRS
12847376	Dual-pathology presacral mass	Meningocele+teratoma in 22/200
30012177	MNX1 in bladder cancer	Overexpression → oncogenesis (allelic series)
38585544	7q36 deletion via t(7;21)	Structural/translocation mechanism
23466002	Variable imaging, 3 siblings	Variable expressivity; MRI family screening

Limitations and Knowledge Gaps

1. **Mutation-negative cases:** ~50% of sporadic CS lacks an identifiable MNX1 variant, indicating unknown modifier genes, regulatory variants, or locus heterogeneity (PMID: 28689883; PMID: 19853743). No modifier gene has been identified.
2. **No genotype–phenotype correlation:** the mechanism underlying variable expressivity and ~50% penetrance is unexplained; stochastic developmental and epigenetic factors are plausible but undemonstrated.
3. **Model-organism mismatch:** the Hlxb9-null mouse best models MNX1 molecular/pancreatic biology, not the human caudal malformation; a heterozygous mouse recapitulating the triad is lacking.
4. **Mechanistic inference:** the caudal dorsal–ventral patterning defect is inferred from gene biology rather than directly demonstrated in human CS tissue; transcriptomic/single-cell data from affected caudal tissue are absent.
5. **Epidemiology:** prevalence is poorly quantified (Orphanet <1/100,000); no population registry or incidence data.
6. **Rare-event surveillance:** the true rate and predictors of malignant transformation are unknown given the disorder's rarity.

Proposed Follow-up Experiments / Actions

1. **Expanded genomic analysis of mutation-negative CS** — trio WGS/RNA-seq to identify non-coding MNX1 regulatory variants, structural variants, and candidate modifier/digenic loci.
2. **Heterozygous and conditional caudal Mnx1 mouse models** — to test whether Mnx1 haploinsufficiency (with sensitized/modifier backgrounds) reproduces the sacral/anorectal/presacral triad and to map the critical developmental window.
3. **Single-cell/spatial transcriptomics of the embryonic caudal eminence** in Mnx1-perturbed models to resolve the inferred dorsal–ventral patterning defect and define affected cell types (notochord, hindgut endoderm, neural tube).
4. **Functional assays of homeobox missense variants** (e.g., p.Arg243Trp) for DNA-binding/transcriptional activity and NONO interaction, to clarify LoF vs altered-specificity mechanisms along the allelic series.

5. **Natural-history registry** — prospective multicenter cohort to quantify penetrance, long-term bowel/bladder outcomes, and malignant-transformation rate, standardizing MRI family-screening protocols.
6. **Standardized cascade-screening guideline** — formalize MRI + MNX1 testing recommendations for first-degree relatives of carriers.

Evidence source types: human clinical (cohorts/case series/reviews), model organism (mouse Hlxb9 knockout), in vitro (SK-N-BE, MIN6). All mechanistic and clinical claims above are cited to primary literature with PMIDs and verbatim abstract quotes where provided in the knowledge base.
